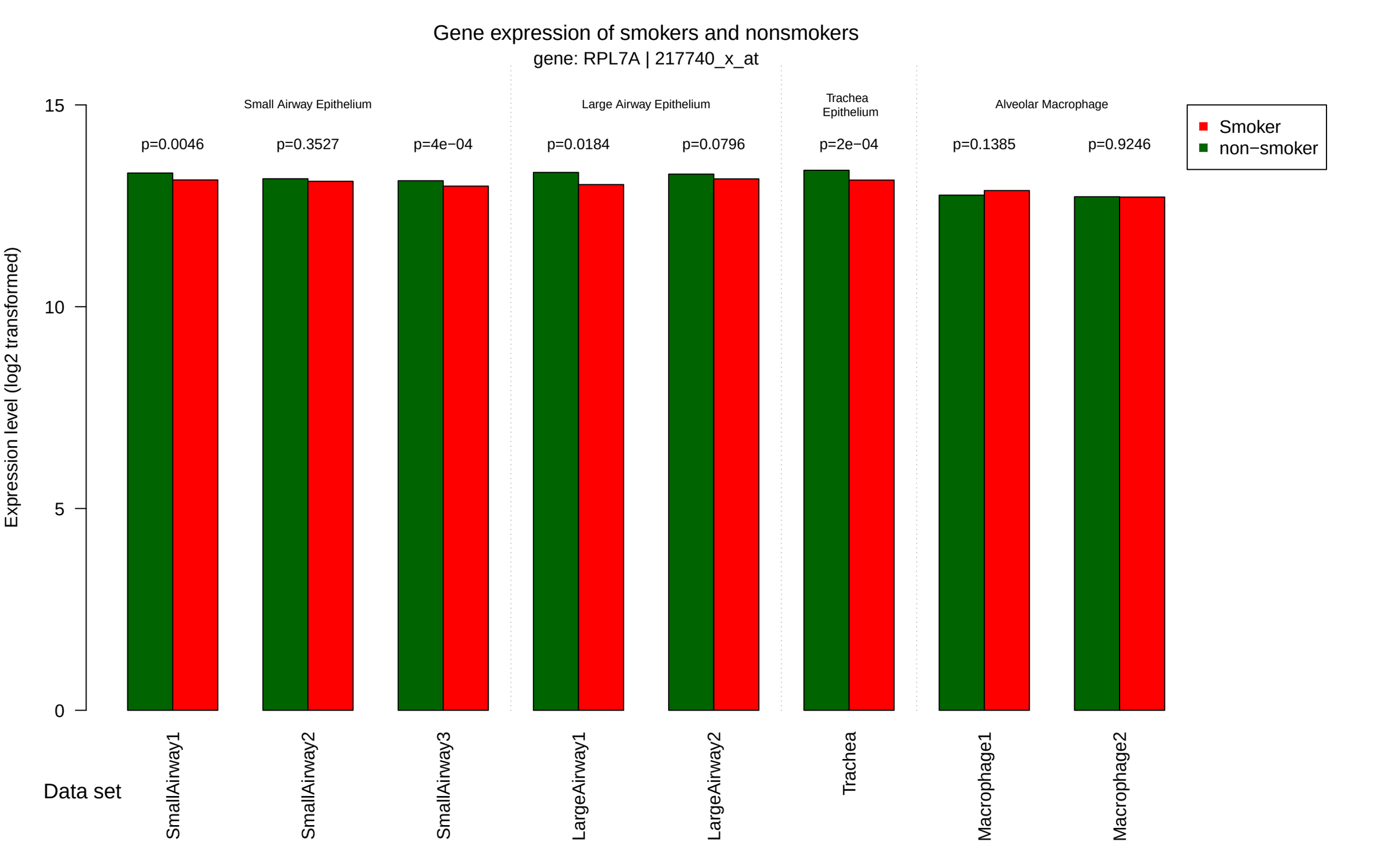
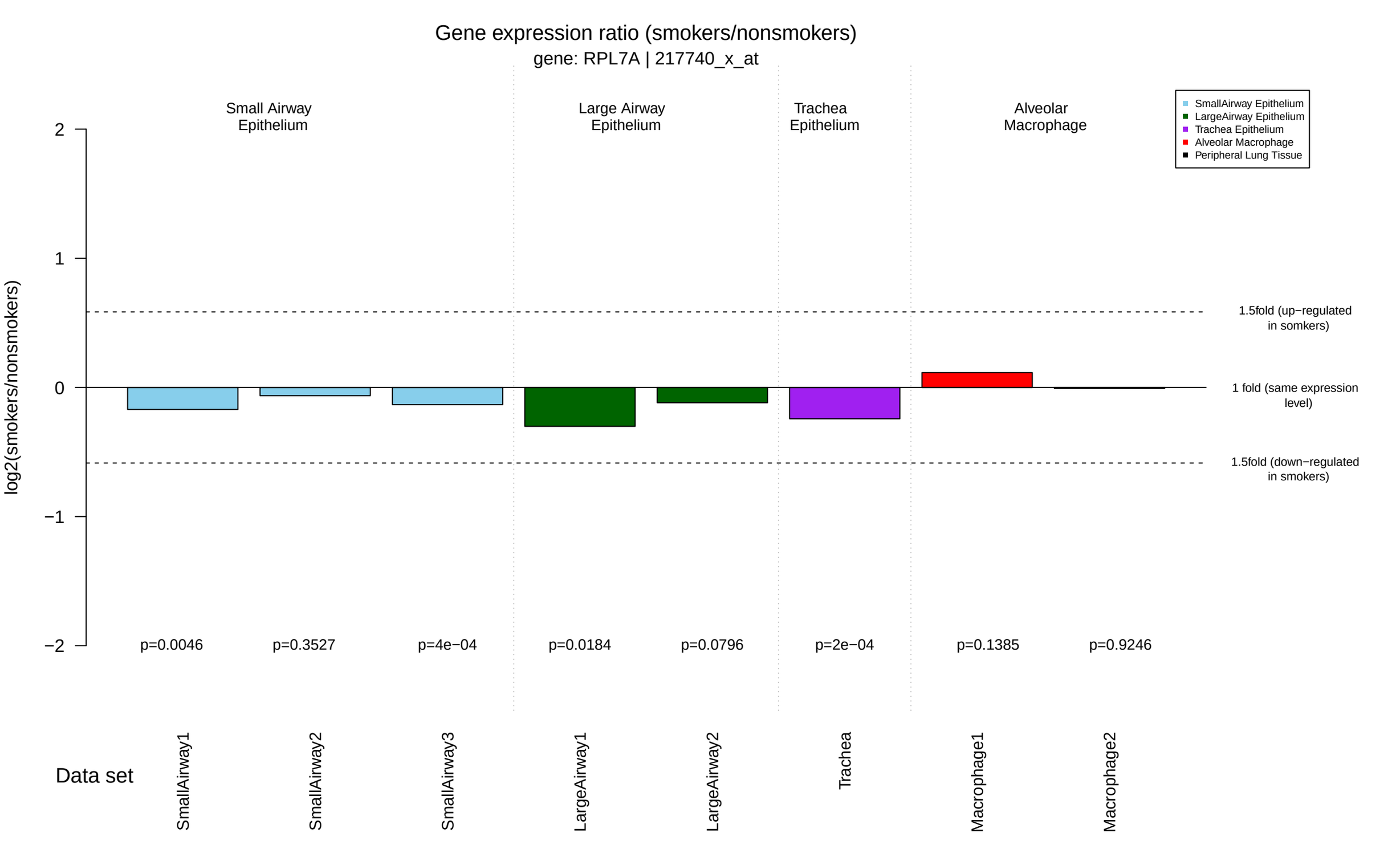
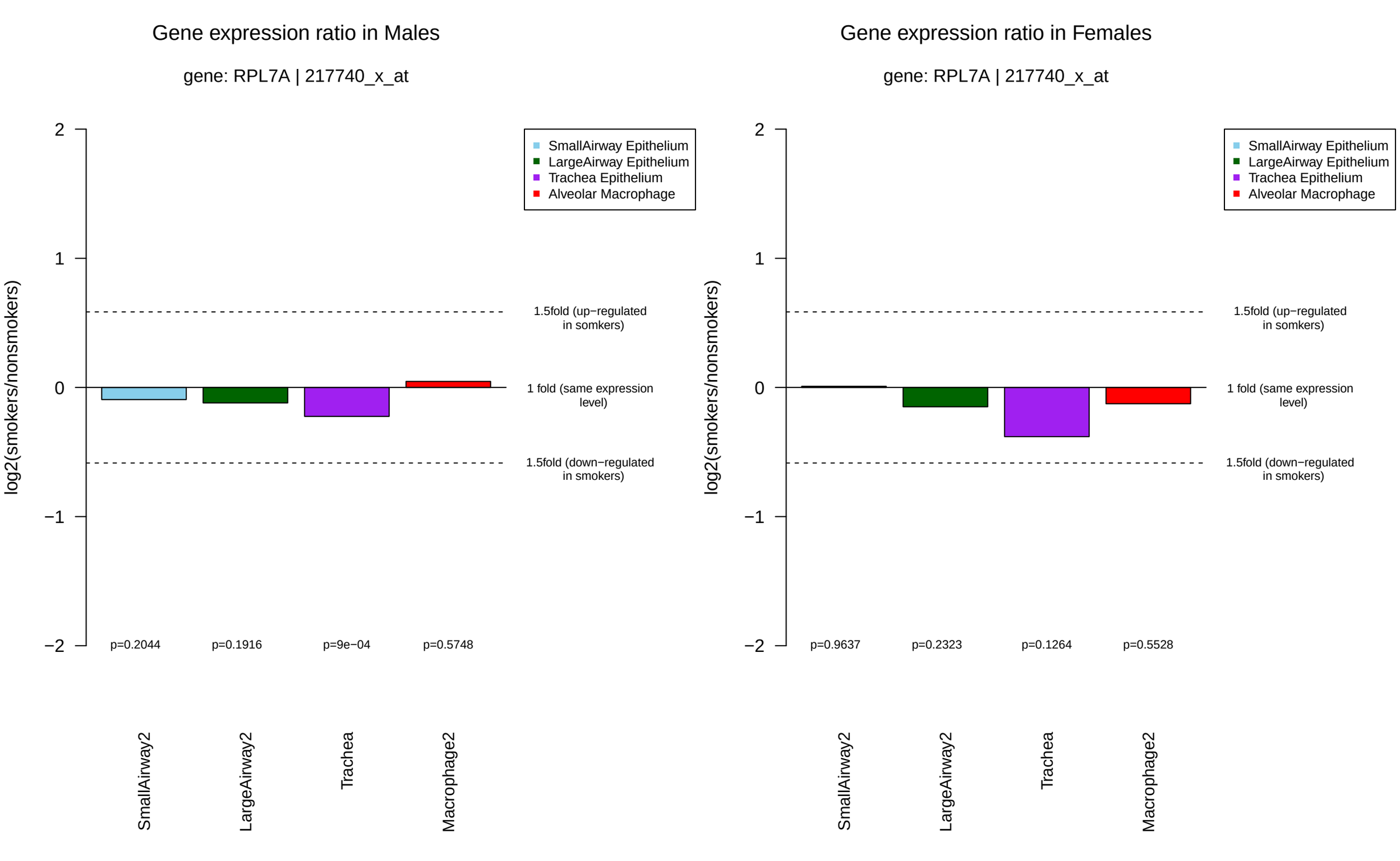
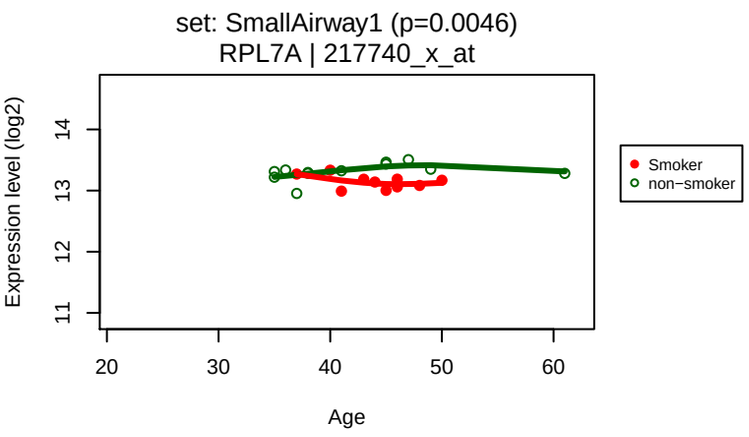
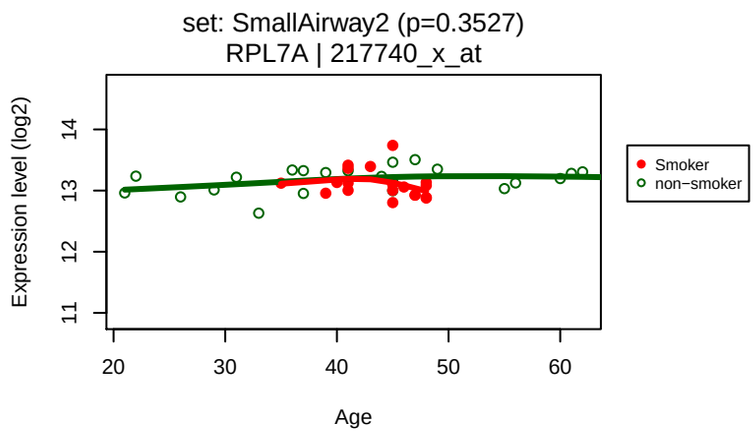
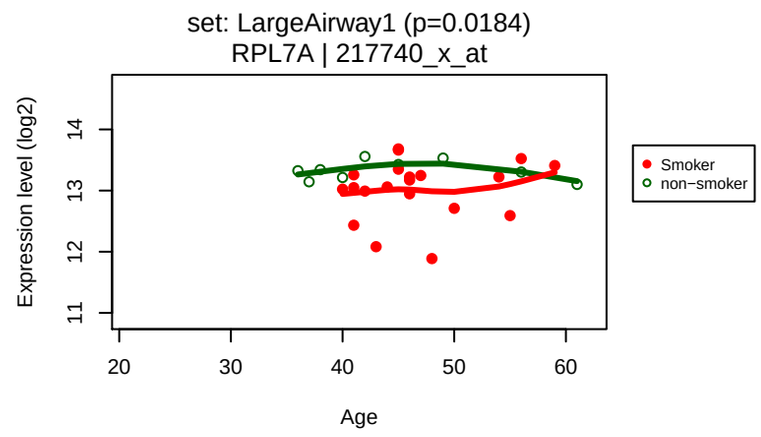

Gene expression of smokers and nonsmokers
gene: RPL7A | 217740_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0046
p=0.3527
p=4e−04
p=0.0184
p=0.0796
p=2e−04
p=0.1385
p=0.9246
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

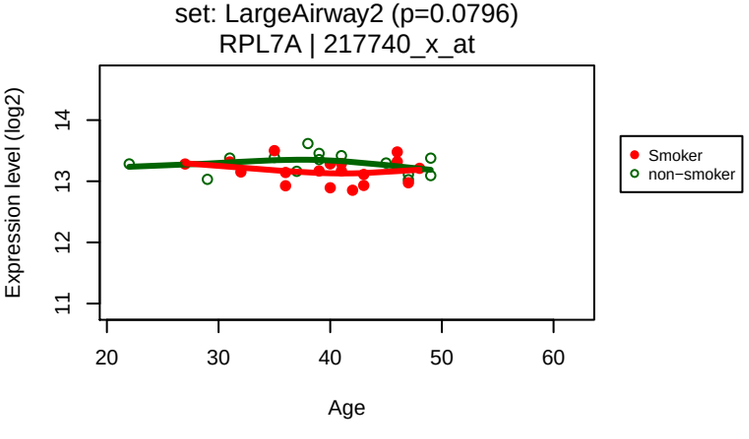

Gene expression ratio (smokers/nonsmokers)
gene: RPL7A | 217740_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0046
p=0.3527
p=4e−04
p=0.0184
p=0.0796
p=2e−04
p=0.1385
p=0.9246
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

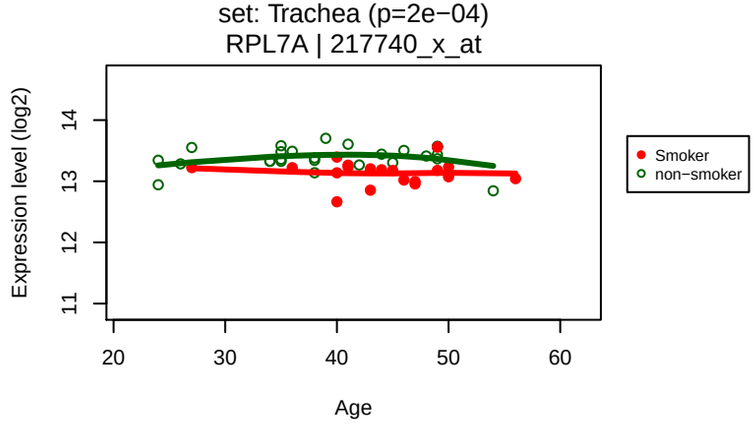

Gene expression ratio in Males
gene: RPL7A | 217740_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2044
p=0.1916
p=9e−04
p=0.5748
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPL7A | 217740_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9637
p=0.2323
p=0.1264
p=0.5528
SmallAirway2
LargeAirway2
Trachea
Macrophage2

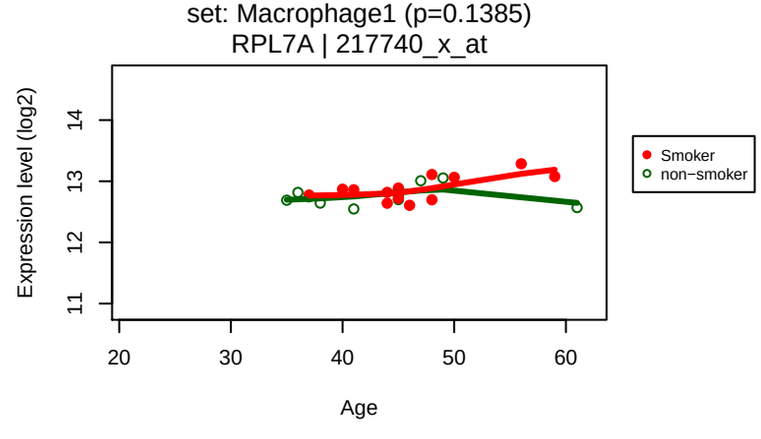

set: SmallAirway1 (p=0.0046)
RPL7A | 217740_x_at
Expression level (log2)
Age
Smoker
non−smoker

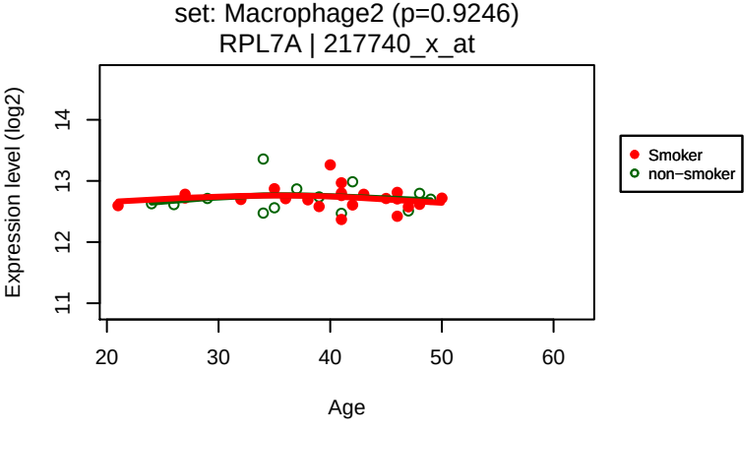

set: SmallAirway2 (p=0.3527)
RPL7A | 217740_x_at
Expression level (log2)
Age
Smoker
non−smoker

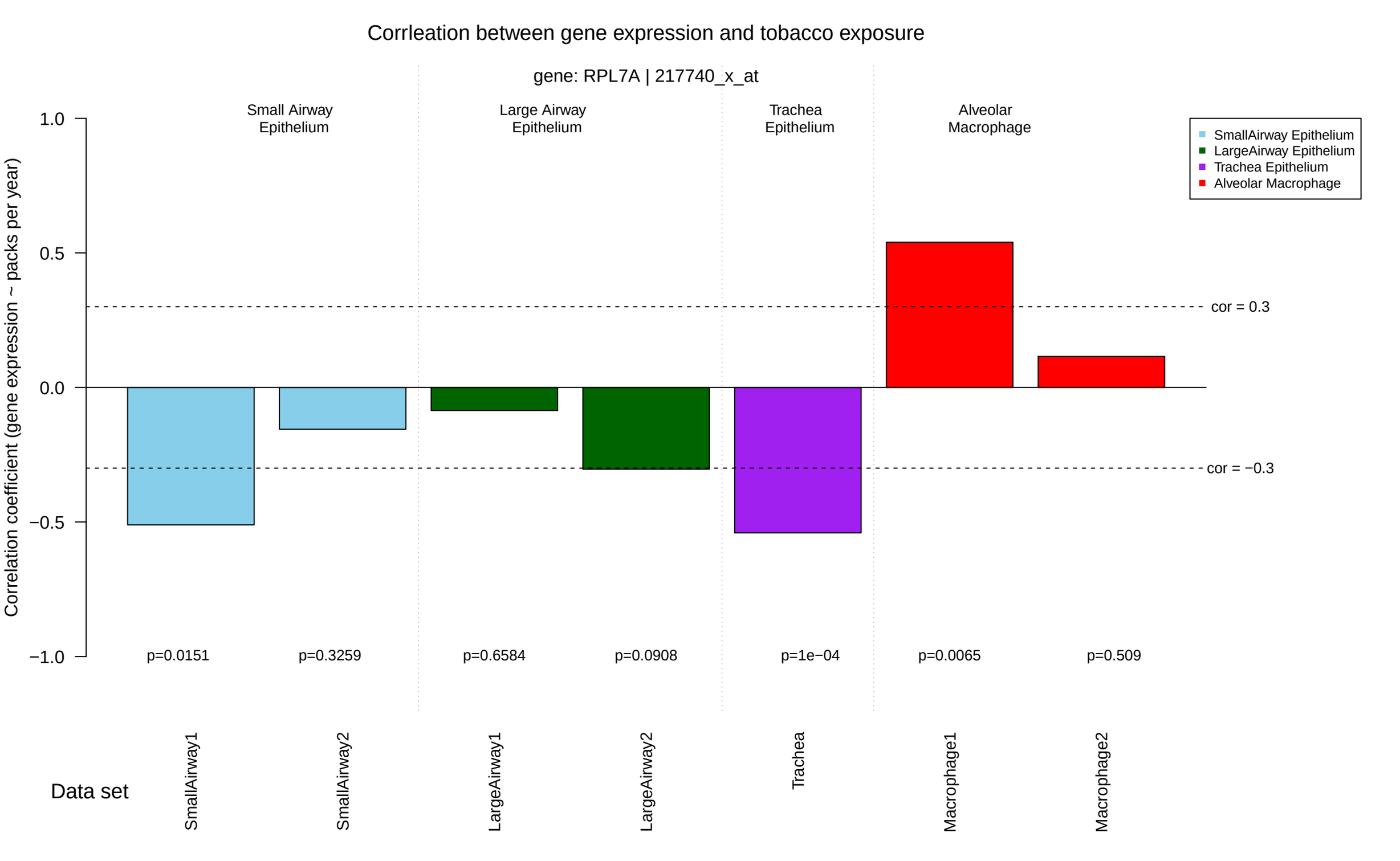

set: LargeAirway1 (p=0.0184)
RPL7A | 217740_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0796)
RPL7A | 217740_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=2e−04)
RPL7A | 217740_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1385)
RPL7A | 217740_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9246)
RPL7A | 217740_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPL7A | 217740_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0151
p=0.3259
p=0.6584
p=0.0908
p=1e−04
p=0.0065
p=0.509
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2